Assignment 3 Report

Jake Weber

1. Network Architecture Framework

The core architecture utilized is the **Mobile Segment Anything Model (MobileSAM)**. Nuclei Instance Segmentation was performed using a Parameter-Efficient Fine-Tuning (PEFT) strategy on a foundation model

- **Image Encoder:** TinyViT
- **Fine-Tuning Strategy:** Low-Rank Adaptation (LoRA)
- **LoRA Injection:** LoRA layers were injected specifically into the Attention blocks of the TinyViT Image Encoder. The linear projection layers for the Query (q) and Value (v) components were wrapped with LoRA matrices. The massive original base model weights were frozen (`requires_grad = False`), ensuring that only the newly injected, low-rank matrices were updated during training.
- **Total Tunable LoRA Parameters:** 29,696

2. Implementation Details

The model was trained and evaluated using the NuInsSeg dataset, utilizing a strict 5-Fold Cross Validation approach to ensure robust generalization across different tissue types.

- **Optimizer:** AdamW
- **Learning Rate:** 1e-4
- **Loss Function:** BCE With Logits Loss. During the training phase, the instance masks were temporarily binarized to calculate the loss against the model's binary logit predictions.
- **LoRA Rank (r):** 4
- **LoRA Alpha (α):** 8
- **Epochs:** 10
- **Batch Size:** 2

- **Data Preparation:** Input images were loaded from the `tissue images` directory. Ground truth instance masks were taken strictly from the `label masks modify` directory. Utilizing the modified instance labels ensured each touching nucleus retained a unique integer ID, which is strictly required for instance-level metric evaluation.

The LoRA layers were applied to all 10 transformer blocks within the MobileSAM framework. This was done to only the transformer blocks because these blocks inherently use 2d matrices to process its data, and LoRA also natively uses 2d matrices to calculate new weights. The ease of matching these made this the proper place to implement this. The transformer blocks are also where the main higher level details and understanding of the image are found, so this is where it is most important to fine tune the model using LoRA. The previous pre-trained weights of the original MobileSAM were frozen, so that the computed LoRA weights were only added on top of the these pre-trained weights, and not replacing the weights.

3. Quantitative Metrics

During the evaluation phase, the network's binary predictions were processed using a connected components algorithm to extract unique instance IDs. These predicted instances were then evaluated against the ground truth `.tif` labels using the Hungarian algorithm for instance matching (IoU threshold > 0.5). The average results across all 5 folds are reported below:

Evaluation Metric	5-Fold Average Score
Average Dice Score (Semantic)	0.6076 (60.76%)
Aggregated Jaccard Index (AJI)	0.4229 (42.29%)
Panoptic Quality (PQ)	0.1405 (14.05%)

Table 3. NuInsSeg segmentation benchmark results based on five-fold cross-validation

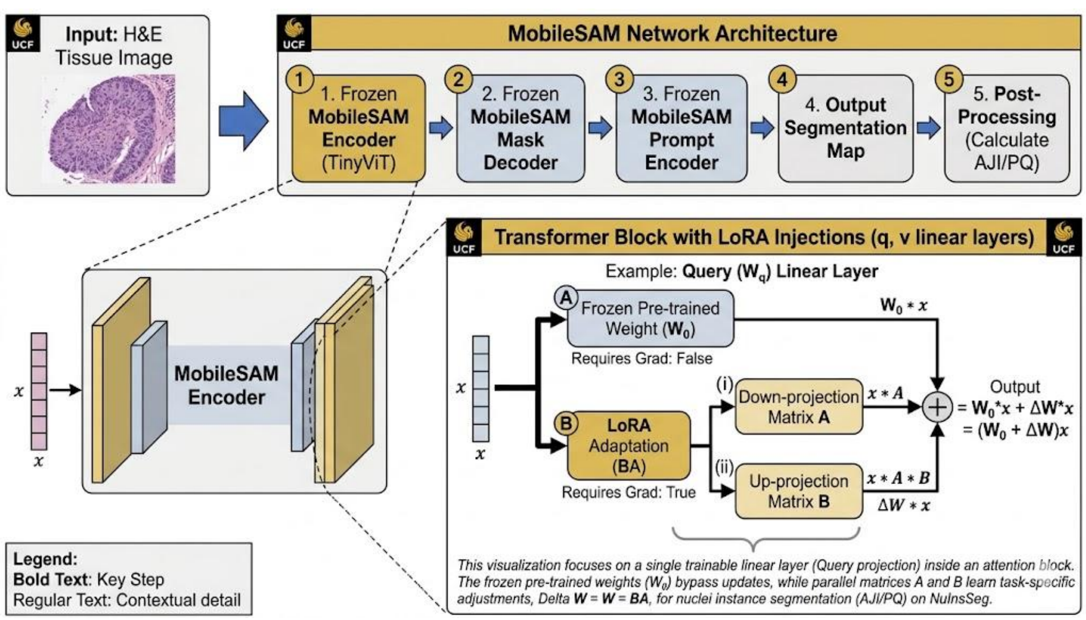
Model	Reference	# Parameters	Avg.Dice (%)	Avg. AJI (%)	Avg. PQ (%)
Shallow U-Net	31	1.9 million	78.8	50.5	42.7
Deep U-Net	31	7.7 million	79.7	49.4	40.4
Attention U-Net	32	2.3 million	80.5	45.7	36.4
Residual attention U-Net	32, 33	2.4 million	81.4	46.2	36.9
Two-stage U-Net	34	3.9 million	76.6	52.8	47.2
Dual decoder U-Net	13	3.5 million	79.4	55.9	51.3

Source: Adapted from Mahbod *et al.* [1].

4. Justifications

All three metrics in my example are lower than results found from all baseline methods from the given paper. The main reason for this is that the model I used was the MobileSAM. This is a highly efficient but less powerful variant of SAM. This was also only trained across 10 epochs. In my testing, the three metrics continued to increase, but were stopped early at epoch 10 due to computation costs and time. Given more epochs and a more powerful model, I believe I could have outperformed the baseline methods.

5. LoRA Layer Injections Framework



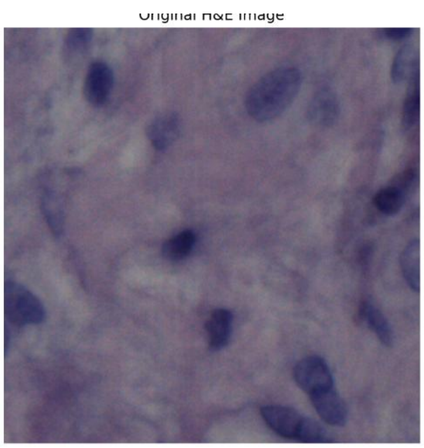
6. Visual Examples

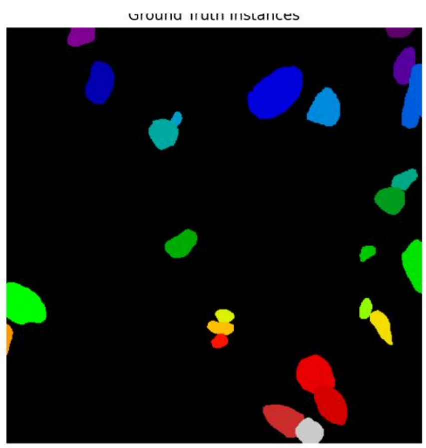
Below are visual comparisons from the final epoch (Epoch 10) of the cross-validation folds, demonstrating the Original H&E Image, the Ground Truth Instances, and the LoRA Prediction.

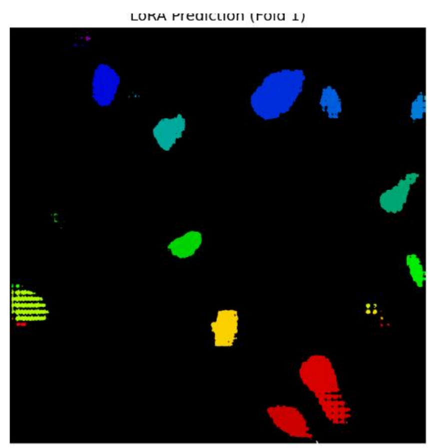
Fold 1

Val Dice: 0.6647 | Val AJI: 0.4658 | Val PQ: 0.1646

Original H&E Image | **Ground Truth Image** | **LoRA Prediction**



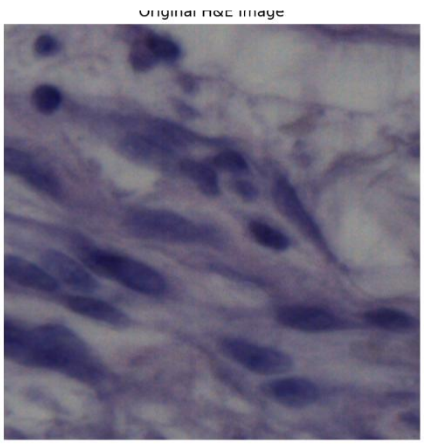
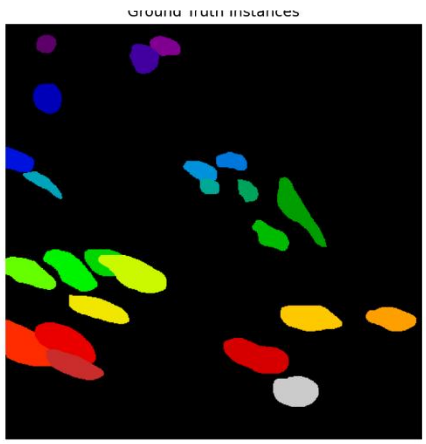
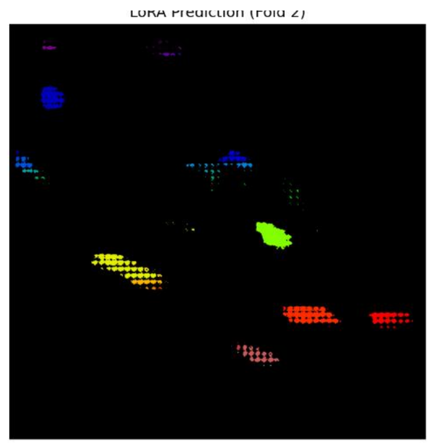




Fold 2

Val Dice: 0.5906 | Val AJI: 0.6535 | Val PQ: 0.1258

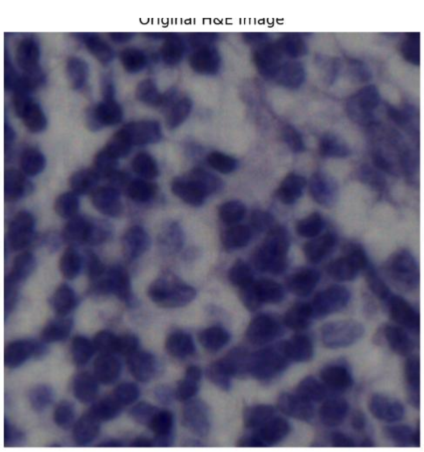
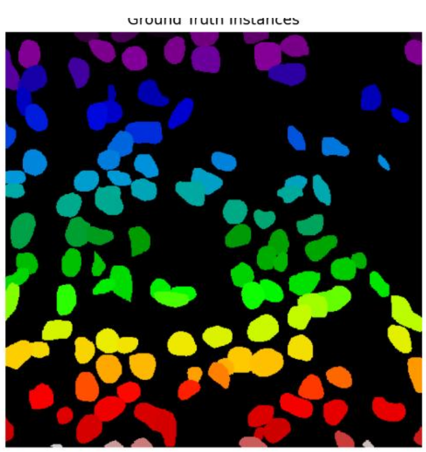
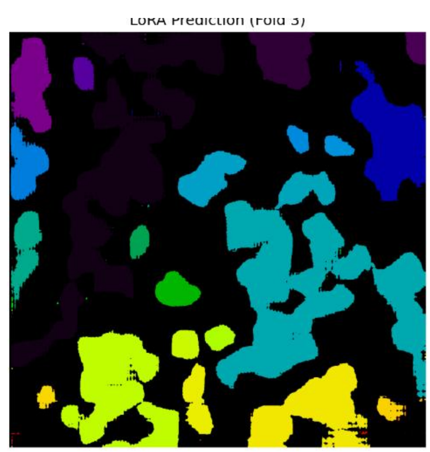
Original H&E Image | **Ground Truth Image** | **LoRA Prediction**

Fold 3

Val Dice: 0.6230 | Val AJI: 0.3485 | Val PQ: 0.1388

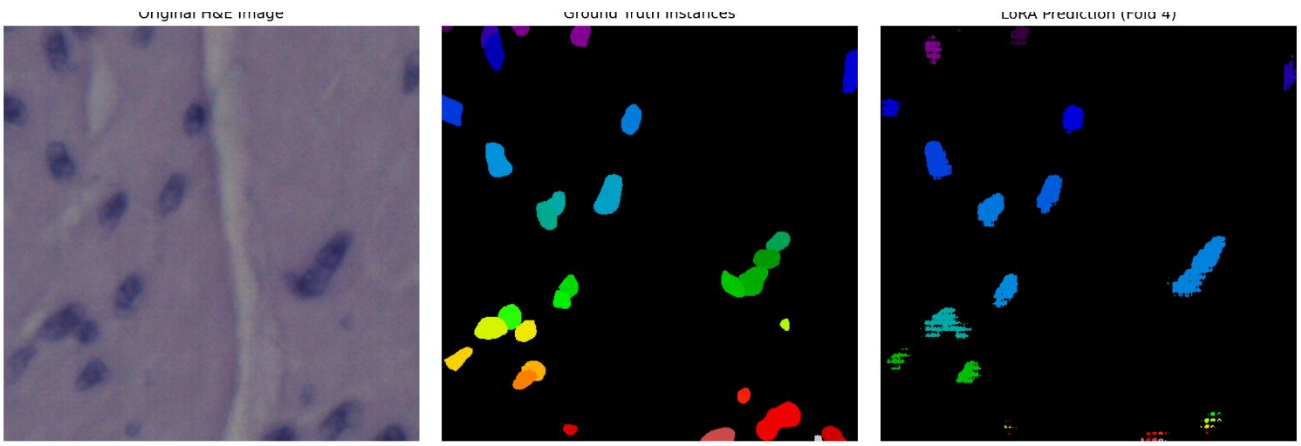
Original H&E Image | **Ground Truth Image** | **LoRA Prediction**

Fold 4

Val Dice: 0.5709 | Val AJI: 0.3352 | Val PQ: 0.1333

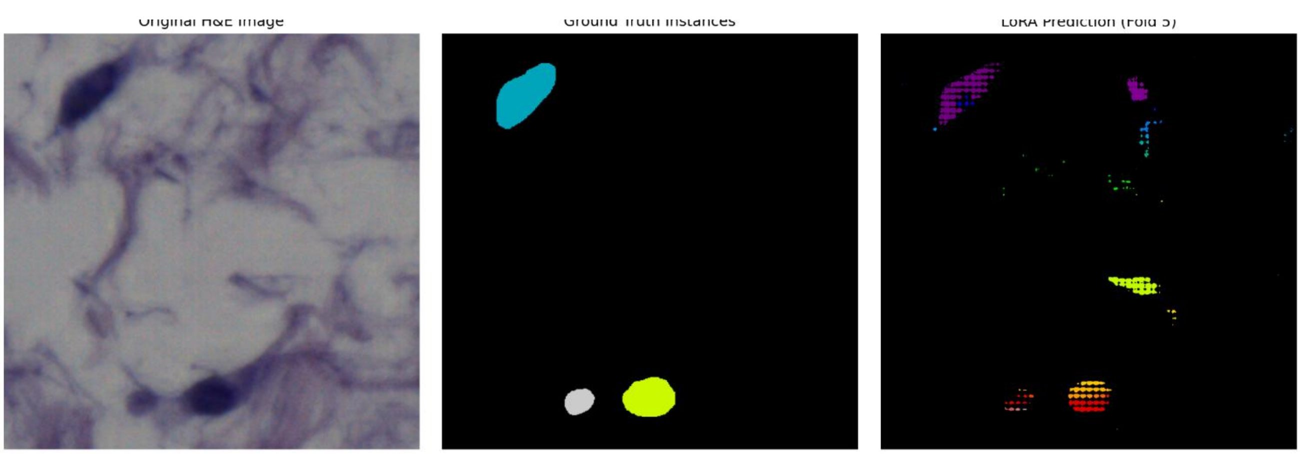
Original H&E Image **Ground Truth Image** **LoRA Prediction**



Fold 5

Val Dice: 0.5888 | Val AJI: 0.3115 | Val PQ: 0.1399

Original H&E Image **Ground Truth Image** **LoRA Prediction**



7. Github

https://github.com/Jeber811/Fine-Tuning-MobileSAM-with-LoRA-for-Nuclei-Instance-Segmentation

8. References

[1] Mahbod, A., Polak, C., Feldmann, K., Khan, R., Gelles, K., Dorffner, G., Woitek, R., Hatamikia, S., & Ellinger, I. (2024). NuInsSeg: A fully annotated dataset for nuclei instance segmentation in H&E-stained histological images. *Scientific Data, 11*(1), 295. https://doi.org/10.1038/s41597-024-03162-w